Abdalrahman Essam Mohamed

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TECHNICAL SKILLS SUMMARY

- **Programming Languages**: Python, C++, Dart, Kotlin, Java, SQL
- **Mobile & Backend:** Flutter, Android, Spring Boot, Node.js, REST APIs, Firebase
- **AI & Data**: PyTorch, Scikit-learn, Pandas, NumPy, NLP, RAG, LLM APIs
- **Core Concepts:** DSA, OOP, Clean Architecture, MVVM, SOLID, Design Patterns, Multithreading
- **Databases:** SQL, Firestore, Database Design

EDUCATION

Bachelor of Communication and Computer Engineering expected 2027 Cairo University, Faculty of Engineering, STEP, Giza, Egypt.

- Major: Communication and Computer Engineering (Computer Track)
- Cumulative GPA: 3.51/4.00

Significant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Software Engineering, Database Management Systems, Computer Networks, Embedded Systems, Pattern Recognition & Neural Networks, Image processing, Natural Language Processing, Generative AI for Software Testing.

Certifications & Specializations:

- ***DeepLearning.AI:*** Natural Language Processing Specialization (Classification, Vector Spaces, Probabilistic Models, Sequence & Attention Models).
- ***Introduction to Model Context Protocol***: Anthropic
- ***ServiceNow*:** IT Automation & AI-Powered Workflows.
- ***Flutter Mobile Application Development*:** Information Technology Institute (ITI)

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- **Web Development Course Creator & Instructor** | Cairo University (CCE Department)
  - Designed and taught a full-stack web curriculum covering networking basics, HTML/CSS/JS, React, database systems, and cloud deployment.
- **Student Representative** | Cairo University CCE Department
  - Communicated student feedback to faculty leadership, resolved academic concerns.
- **GEN-AI member at CURT** (Cairo University racing team)

EXPERIENCE

Data Engineering Intern | Summit Technology Solutions
2026-08-09 – 2026-09-06

- Built ETL/ELT pipelines using Python, dbt, and Snowflake.
- Applied data warehousing, data modeling, and data visualization using Power BI.
- Developed an end-to-end data engineering project from data ingestion to dashboard creation.

Network engineer Intern | Summit Technology Solutions

2026-08-05 – 2026-09-0

- Studied enterprise networking, including VLANs, routing, wireless networks, and Wireless LAN Controllers (WLCs).
- Gained exposure to Cisco Nexus, FTD, SDA, and SD-WAN technologies.
- Visited the company's data center to explore enterprise network infrastructure and architecture.

Software engineer Intern | Summit Technology Solutions

2026-07-05 – 2026-08-03

- Developed native Android applications using **Kotlin** following modern Android development practices.
- Integrated AI-powered features to enhance application functionality and user experience.

ServiceNow Developer Intern | Ejada

2026-07-05 – 2026-08-02

- Participating in a structured ServiceNow developer training program.
- Learning App Engine Studio, Flow Designer, Reports, UI Builder, Service Portal, and platform scripting.
- Developing hands-on applications and automation workflows in a project called **TrackFlow**.
- Working with mentors to gain practical experience in enterprise application development.

Flutter Development Intern | ET3 - Tomorrow Information Technology

2025-06-24 – 2025-07-15

- Contributed to the development of "**Sabeqoon**," a comprehensive school management mobile application using Flutter.
- **Developed and implemented the translation** mechanism for three languages (Arabic, English, Turkish)
- Translated UI/UX designs into responsive and high-performance mobile screens using Flutter's widget-based architecture.

Projects Worked On:

Industrial Machine Sound Classification & Anomaly Detection | GitHub · Hugging Face

- Built an audio preprocessing pipeline and a 2D CNN (2.49M parameters) for machine sound classification using log-mel spectrograms.
- **Technologies: Python, PyTorch, Librosa, Scikit-learn.**

Multilingual News Intelligence Engine | GitHub · Hugging Face

- Built a multilingual news pipeline processing 45+ feeds with topic classification, sentiment analysis, and named entity recognition.
- Technologies: Python, PyTorch, Hugging Face, Firebase.

Decibel Audio Streaming Platform | GitHub

- Led a Flutter development team, implementing Clean Architecture, authentication, and social features.
- Technologies: Flutter, Dart, Riverpod, GoRouter, Dio.

Autonomous Wall-Following Robot | C GitHub

- Developed low-level hardware drivers for sensors and actuators.

Tetris War | C#, Unity GitHub

- Developed a Tetris-inspired strategy game combining block placement with army-building mechanics.

OS Scheduler

- Designed and implemented a process scheduler in C on Linux, managing CPU time allocation and process execution using multiple scheduling algorithms.